1 Multivarite Distribution

1.1 Transformations of Several Random Variables

Exercise 1.1.1. Let X_1, X_2, X_3 be iid each with distribution having pdf $f(x) = e^{-x}$, $x \in \mathbb{R}^+$. Show that:

$$Y_1 = \frac{X_1}{X_1 + X_2} \quad Y_2 = \frac{X_1 + X_2}{X_1 + X_2 + X_3} \quad Y_3 = X_1 + X_2 + X_3$$

are mutually independent.

Solution,

By independence, the joint pdf is:

$$f(x_1, x_2, x_3) = e^{-x_1 - x_2 - x_3} \quad (x_1, x_2, x_3) \in \prod_{n=1}^{3} \mathbb{R}^+$$

Now, the inverse transforms are:

$$\begin{aligned} X_1 &= Y_1 Y_2 Y_3 \\ X_2 &= (1 - Y_1)(Y_2 Y_3) \\ X_3 &= (1 - Y_2) Y_3 \end{aligned}$$

The Jacobian is:

$$J = \begin{pmatrix} Y_1 Y_3 & Y_1 Y_3 & Y_1 Y_2 \\ -Y_2 Y_3 & Y_3(1 - Y_1) & Y_2(1 - Y_1) \\ 0 & -Y_3 & 1 - Y_2 \end{pmatrix}$$

Therefore, the Jacobian determinant is:

$$\det J = \det \begin{pmatrix} Y_1 Y_3 & Y_1 Y_3 & Y_1 Y_2 \\ -Y_2 Y_3 & Y_3(1 - Y_1) & Y_2(1 - Y_1) \\ 0 & -Y_3 & 1 - Y_2 \end{pmatrix} = Y_2 Y_3^2$$

So, the pdf of (Y_1, Y_2, Y_3) is:

$$f_{Y_1, Y_2, Y_3}(y_1, y_2, y_3) = |\det J| e^{-y_3} = y_2 y_3^2 e^{-y_3}$$

Now, over the support of (Y_1, Y_2, Y_3) the pdf is indeed a product of functions of y_1, y_2, y_3. However, for mutual independence, the support must also be a product space.

From the inverse transforms, namely X_2, X_3, we have the constraint:

$$y_1, y_2 \in (0, 1)$$

Similarly,

$$Y_3 = X_1 + X_2 + X_3 \implies y_3 \in (0, \infty)$$

The equations don't constraint the variables further, thus, we get:

$$(y_1, y_2, y_3) \in (0, 1) \times (0, 1) \times (0, \infty)$$

Thus, (Y_1, Y_2, Y_3) is linearly independent.

□

Exercise 1.1.2. If $f(x) = \frac{1}{2}, x \in (-1, 1)$ is the pdf of the random variable X, find the pdf of $Y = X^2$

Solution,

Here, the issue is that Y is not an injective transformation. Fortunately, $(0, 1) \xrightarrow{Y} (0, 1)$ and $(-1, 0) \xrightarrow{Y} (0, 1)$.

Therefore,

$$f_Y(y) = f_X(\sqrt{y}) \left| \frac{\mathrm{d}}{\mathrm{d}y}(\sqrt{y}) \right| + f_X(-\sqrt{y}) \left| \frac{\mathrm{d}}{\mathrm{d}y}(-\sqrt{y}) \right| = \frac{1}{2} \left(\frac{1}{2\sqrt{y}} + \frac{1}{2\sqrt{y}} \right)$$

$$\therefore f_Y(y) = \frac{1}{2\sqrt{y}}$$

Now, for $y = 0$,

$$f_Y(0) = 0$$

Although, we are free to re-define to any value we wish to in $\mathbb{R}$ or just leave it be. Defining it to be zero makes the density nice.

Thus, we get the density:

$$f_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}} & y \in (0,1) \\ 0 & \text{elsewhere} \end{cases}$$

□

Exercise 1.1.3. If X has the pdf of $f(x) = \frac{1}{4}, -1 < x < 3$, zero elsewhere, find the pdf of $Y = X^2$

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Solution,

We can break:

$$(-1, 3)$$

into:

$$(-1, 0) \cup (0, 1) \cup (1, 3)$$

$Y = X^2$ transforms $(-1, 0) \cup (0, 1) \to (0, 1)$. So, for $y \in (0, 1)$, the density follows that:

$$f_Y(y) = f(-\sqrt{y}) \left| \frac{\mathrm{d}}{\mathrm{d}y}(-\sqrt{y}) \right| + f(\sqrt{y}) \frac{\mathrm{d}}{\mathrm{d}y}(\sqrt{y}) = \frac{1}{4} \left(\frac{1}{2\sqrt{y}} + \frac{1}{2\sqrt{y}} \right) = \frac{1}{4\sqrt{y}}$$

Now, for $y \in (1, 9)$:

$$f_Y(y) = f(\sqrt{y}) \frac{\mathrm{d}}{\mathrm{d}y}(\sqrt{y}) = \frac{1}{8\sqrt{y}}$$

Therefore, it follows the density of Y is given by:

$$f_Y(y) = \begin{cases} \frac{1}{4\sqrt{y}} & y \in (0,1) \\ \frac{1}{8\sqrt{y}} & y \in (1,9) \\ 0 & \text{elsewhere} \end{cases}$$

□

Exercise 1.1.4. Let X_1, X_2, X_3 be iid with common pdf $f(x) = \exp(-x), x > 0, 0$ elsewhere. Find the joint pdf of $Y_1, = X_1, Y_2 = X_1 + X_2, Y_3 = X_1 + X_2 + X_3$.

Solution,

By independence,

$$f(x_1, x_2, x_3) = \begin{cases} e^{-(x_1+x_2+x_3)} & (x_1, x_2, x_3) \in \prod_{n=1}^{3} \mathbb{R}^+ \\ 0 & \text{elsewhere} \end{cases}$$

The inverse transforms is:

$$\begin{aligned} X_1 &= Y_1 \\ X_2 &= Y_2 - Y_1 \\ X_3 &= Y_3 - Y_2 \end{aligned}$$

By positivity, it follows that:

$$0 < Y_1 < Y_2 < Y_3 \quad Y_1 \in (0, \infty)$$

Therefore, the support of Y is:

$$\mathcal{S}_Y = \{(y_1, y_2, y_3) \in \mathbb{R}^3 : 0 < y_1 < y_2 < y_3 < \infty\}$$

The Jacobian is:

$$J = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \implies \det J = 1$$

Hence, the pdf is:

$$f_Y(y_1, y_2, y_3) = \begin{cases} e^{-y_3} & 0 < y_1 < y_2 < y_3 < \infty \\ 0 & \text{elsewhere} \end{cases}$$

□

Exercise 1.1.5. Let X_1, X_2, X_3 be iid with common pdf $f(x) = e^{-x}, x > 0, 0$ elsewhere. Find the joint pdf of:

$$Y_1 = \frac{X_1}{X_2} \quad Y_2 = \frac{X_3}{X_1 + X_2} \quad Y_3 = X_1 + X_2$$

Are Y_1, Y_2, Y_3 independent?

Solution,

By independence,

$$f(x_1, x_2, x_3) = \begin{cases} e^{-x_1 - x_2 - x_3} & x_1, x_2, x_3 \in \mathbb{R}^+ \\ 0 & \text{elsewhere} \end{cases}$$

The inverse transforms are:

$$X_1 = \frac{Y_1 Y_3}{1 + Y_1} \quad X_2 = \frac{Y_3}{1 + Y_1} \quad X_3 = Y_2 Y_3$$

Hence, the Jacobian is:

$$J = \begin{pmatrix} \frac{Y_3}{(Y_1+Y_2)^2} & 0 & \frac{Y_1}{Y_1+1} \\ -\frac{Y_3}{(Y_1+1)^2} & 0 & \frac{1}{Y_1+1} \\ 0 & Y_3 & Y_2 \end{pmatrix} \implies \det J = -\frac{Y_3^2}{(Y_1+1)^2}$$

Therefore,

$$|\det J| = \left(\frac{Y_3}{1 + Y_1}\right)^2$$

So, over $\mathcal{S}_Y$, the joint pdf is:

$$f_Y(y_1, y_2, y_3) = \left(\frac{y_3}{1 + y_1}\right)^2 y_3(1 + y_2) = \frac{y_3^3(1 + y_2)}{(1 + y_1)^2}$$

Finally, we figure out $\mathcal{S}_Y$.

There are no constraint which the transformation forces. So, we can conclude that $y_1, y_2, y_3 \in \mathbb{R}^+$.

Thus,

$$f_Y(y_1, y_2, y_3) = \begin{cases} \frac{y_3^2(1+y_2)}{(1+y_1)^2} & y_1, y_2, y_3 \in \mathbb{R}^+ \\ 0 & \text{elsewhere} \end{cases}$$

Also, (Y_1, Y_2, Y_3) are mutually independent.

□

Exercise 1.1.6. Let X_1, X_2 have the joint pdf:

$$f(x_1, x_2) = \begin{cases} \frac{1}{\pi} & 0 < x_1^2 + x_2^2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Let $Y_1 = X_1^2 + X_2^2, Y_2 = X_2$. Find the joint pdf of Y_1, Y_2

Solution,

Unfortunately, the inverse transform here isn't injective. In fact,

$$X_1 = \pm\sqrt{Y_1 - Y_2^2} \quad X_2 = Y_2$$

Therefore, we have to compute two Jacobians and sum them together.

Now, we've two Jacobians,

$$J_1 = \begin{pmatrix} \frac{1}{2\sqrt{Y_1 - Y_2^2}} & -\frac{Y_2}{\sqrt{Y_1 - Y_2^2}} \\ 0 & 1 \end{pmatrix} \quad J_2 = \begin{pmatrix} -\frac{1}{2\sqrt{Y_1 - Y_2^2}} & \frac{Y_2}{\sqrt{Y_1 - Y_2^2}} \\ 0 & 1 \end{pmatrix}$$

Thankfully, the absolute value of the determinant of both of these expressions is:

$$|\det J| = \frac{1}{2\sqrt{Y_1 - Y_2^2}}$$

Hence, the pdf:

$$f_Y(y_1, y_2) = \frac{1}{2\sqrt{y_1 - y_2^2}}\left(\frac{1}{\pi} + \frac{1}{\pi}\right) = \frac{1}{\pi\sqrt{y_1 - y_2^2}}$$

Of course, we have unique inverse when $X_1 = 0 \implies Y_1 = Y_2^2$, in which case:

$$f_Y(y_1, y_2) = \frac{1}{2\pi\sqrt{y_1 - y_2^2}}$$

But of course, the set where it's inverse is unique is of measure zero, so we can ignore its "contribution" to the density.

Now,

$$\mathcal{S}_Y = \{(y_1, y_2) \in \mathbb{R}^2 : y_1 \in (0,1), y_2 \in (-\sqrt{y_1}, \sqrt{y_1})\}$$

The fact that y_2 is not strict is thanks to us ignoring the region of unique inverse.

Finally, putting the both together we get:

$$f_Y(y_1, y_2) = \begin{cases} \frac{1}{\pi\sqrt{y_1^2 - y_2^2}} & (y_1, y_2) \in \mathcal{S}_Y \\ 0 & \text{elsewhere} \end{cases}$$

□

Exercise 1.1.7. Let X_1, X_2, X_3 be iid with common mgf $M(t) = \left(\frac{3}{4} + \frac{1}{4}e^t\right)^2$ for all $t \in \mathbb{R}$.

(a) Determine the probabilities $\mathbb{P}(X_1 = k)$ for $k = 0, 1, 2$.

(b) Find the mgf of $Y = X_1 + X_2 + X_3$, and then determine the probabilities, $\mathbb{P}(Y = k)$

Solution,

Here,

$$M(t) = \left(\frac{3}{4} + \frac{1}{4}e^t\right)^2 = \left(\frac{3}{4}\right)^2 + 2\left(\frac{3}{4}\right)\left(\frac{1}{4}\right)e^t + \left(\frac{1}{4}\right)^2 e^{2t}$$

Therefore,

$$\mathbb{P}(X = 0) = \frac{9}{16} \quad \mathbb{P}(X = 1) = \frac{3}{8} \quad \mathbb{P}(X = 2) = \frac{1}{16}$$

By independence,

$$M_Y(t) = \left(\frac{3}{4} + \frac{1}{4}e^t\right)^6$$

like above, expand the term using the Binomial theorem and match coefficients.

□

1.2 Linear Combinations of Random Variables

Definition 1.2.1 (Linear Combination of Random Variables): Let $A = \begin{pmatrix} X_1 & \dots & X_n \end{pmatrix}^\top$ denote a random variable. A linear combination of these variable is:

$$T = \sum_{i=1}^{n} a_i X_i = a^\top X$$

where, $a = (a_i) \in \mathbb{R}^n$.

□

Theorem 1.2.1 (Expectation of a Linear Combination): Let X be a n-dimensional random vector and $a \in \mathbb{R}^n$, such that $T = a^\top X$. Then,

$$\mathbb{E}T = a^\top \mathbb{E}X$$

Proof,

This trivially follows expectations on random matrices and vectors.

□

Theorem 1.2.2 (Covariance of Linear Combination): Let X, Y be an n-dimensional, m-dimensional random vector resp. Suppose T, W are two linear combinations:

$$T = a^\top X \quad W = b^\top Y \quad a, b \in \mathbb{R}^n, \mathbb{R}^m \text{ resp.}$$

provides the relevant expectations exist (namely the second moment), then:

$$\text{cov}(T, W) = \mathbb{E}(T - \mathbb{E}T)(W - \mathbb{E}W) = a^\top \Sigma_{XY} b$$

where,

$$\Sigma_{XY} = \text{cov}(X, Y) = \mathbb{E}(X - \mathbb{E}X)(Y - \mathbb{E}Y)^\top$$

Proof,

Recall that,

$$\mathbb{E}T = a^\top \mathbb{E}X \implies T - \mathbb{E}T = a^\top X - a^\top \mathbb{E}X = a^\top (X - \mathbb{E}X)$$

Similarly,

$$W - \mathbb{E}W = b^\top (Y - \mathbb{E}Y)$$

However, because, $W - \mathbb{E}W$ is a scalar, so it is equal to its own transpose, hence,

$$W - \mathbb{E}W = (Y - \mathbb{E}Y)^\top b$$

So,

$$\mathbb{E}(T - \mathbb{E}T)(W - \mathbb{E}W) = \mathbb{E}a^\top (X - \mathbb{E}X)(Y - \mathbb{E}Y)^T b = \mathbb{E}a^\top \Sigma_{XY} b$$

Since, $a^\top \Sigma_{XY} b$ is a constant, expectation acts as an identity. Thus,

$$\text{cov}(T, W) = a^\top \Sigma_{XY} b$$

□

Remark: Expanding out we get:

$$\operatorname{cov}(T, W) = \sum_{i=1}^{n} \sum_{j=1}^{m} a_i b_j \operatorname{cov}\left(X_i, Y_j\right)$$

□

Corollary 1.2.2.1: Let $T = a^\top X$ where, $a \in \mathbb{R}^n$ and X is an n-dimensional random vector.

$$\operatorname{var}(T) = \operatorname{cov}(T, T) = a^\top \Sigma_T a$$

□

Corollary 1.2.2.2: If $X_1, ..., X_n$ are mutually independent random variable, then it follows:

$$\operatorname{var}(T) = \operatorname{var}\left(a^\top T\right) = a^\top \operatorname{diag}(\operatorname{var} X_i) a$$

□

Definition 1.2.2 (Random sample and iid): If the random variable $X_1, ..., X_n$ are independent and identically distributed then we say that these random variables constitute a random sample of size n from that common distribution. We abbreviate independent and identically distributed by iid.

□

Example (Sample Mean): Let $X_1, ..., X_n$ be independent and identically distributed variable with common mean μ and variance σ^2. The sample mean is defined to be:

$$\overline{X} = n^{-1} \sum_{i=1}^{n} X_i$$

Thus,

$$\overline{X} = \frac{1}{n} \mathbf{1}^\top X$$

Hence,

$$\mathbb{E}\overline{X} = \frac{1}{n} \mathbf{1}^\top \mathbb{E}X = \frac{n\mathbb{E}X}{n} = \mathbb{E}X$$

and,

$$\begin{aligned}
\operatorname{var} \overline{X} &= \left(\frac{1}{n} \mathbf{1}^\top \right) \operatorname{diag}(\operatorname{var} X)_i \left(\frac{1}{n} \mathbf{1} \right) \\
&= \frac{1}{n^2} \mathbf{1}^\top (\operatorname{var} X) \mathbf{1} \\
&= \frac{\operatorname{var} X}{n^2} (\mathbf{1}^\top \mathbf{1}) \\
&= \frac{\operatorname{var} X}{n^2} \times n \\
&= \frac{\operatorname{var} X}{n}
\end{aligned}$$

□

Example (Sample Variance): We define the sample variance by:

$$S^2 = (n-1)^{-1} \sum_{i=1}^{n} \left(X_i - \overline{X} \right)^2$$

By simplifying and then taking expectations, we get:

$$\mathbb{E}S^2 = \operatorname{var} X$$

□

Exercise 1.2.1. If $X_1, ..., X_4$ be four iid random variables having the pdf:

$$f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find the mean and variance of the sum Y of these four random variables.

Solution,

Since,

$$Y = X_1 + ... + X_4 \implies \mathbb{E}Y = 4\mathbb{E}X$$

Now,

$$\mathbb{E}X = \int_0^1 \mathrm{d}x \, x(2x) = \frac{2}{3}$$

Therefore,

$$\mathbb{E}Y = 4\left(\frac{2}{3}\right) = \frac{8}{3}$$

Similarly,

$$\operatorname{var} Y = (1 \ \ ... \ \ 1)^\top \Sigma_X \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$$

by independence,

$$\operatorname{var} Y = 4 \operatorname{var} X$$

So,

$$\operatorname{var} X = \int_0^1 \mathrm{d}x \, x^2(2x) - \left(\frac{2}{3}\right)^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$$

Hence,

$$\operatorname{var} Y = 4\left(\frac{1}{18}\right) = \frac{2}{9}$$

□

Exercise 1.2.2. Let X_1, X_2be two independent random variables so that the variances of X_1, X_2 are $\sigma_1^2 = k, \sigma_2^2 = 2$, respectively. .Given that the variance of $Y = 3X_2 - X_1$ is 25, find k.

Solution,

Since, X_1, X_2 are independent:

$$\operatorname{var} Y = 3 \operatorname{var} X_2 - \operatorname{var} X_1 \implies 25 = 3(2) - k$$

Thus,

$$k = 6 - 25 = -19$$

□

Exercise 1.2.3. If the independent variables X_1, X_2 have means μ_1, μ_2 and variances σ_1^2, σ_2^2 respectively, show that the mean and variances of the product $Y = X_1 X_2$ are $\mu_1 \mu_2$ and $\sigma_1^2 \sigma_2^2 + \mu_1^2 \sigma_2^2 + \mu_2^2 \sigma_1^2$ respectively.

Solution,

By independence,

$$\mathbb{E}(Y) = \mathbb{E}X_1 \mathbb{E}X_2 = \mu_1 \mu_2$$

Thus,

$$\begin{aligned}
\operatorname{var}(Y) &= \mathbb{E}(X_1 X_2)^2 - \mu_1^2 \mu_2^2 \\
&= \mathbb{E}X_1^2 \mathbb{E}X_2^2 - \mu_1^2 \mu_2^2 \\
&= (\sigma_1^2 + \mu_1^2)(\sigma_2^2 + \mu_2^2) - \mu_1^2 \mu_2^2 \\
&= \sigma_1^2 \sigma_2^2 + \mu_1^2 \sigma_2^2 + \mu_2^2 \sigma_1^2
\end{aligned}$$

□

Exercise 1.2.4. Find the mean and variance of the sum $Y = \sum_{i=1}^{5} X_i$ where X_i are iid having pdf:

$$f(x) = \begin{cases} 6x(1-x) & x \in (0,1) \\ 0 & \text{elsewhere} \end{cases}$$

Solution,

Now,

$$\mathbb{E}X = \int_0^1 \mathrm{d}x\,(x)(6x(1-x)) = \int_0^1 \mathrm{d}x\,6x^2(1-x) = \frac{1}{2}$$

Similarly,

$$\mathbb{E}X^2 = \int_0^1 \mathrm{d}x\,6x^3(1-x) = \frac{3}{10}$$

Thus,

$$\operatorname{var} X = \frac{3}{10} - \left(\frac{1}{2}\right)^2 = \frac{1}{20}$$

So,

$$\mathbb{E}Y = 5\left(\frac{1}{2}\right) = \frac{5}{2}$$

By linearity, the variance is:

$$\operatorname{var} Y = \frac{5}{20} = \frac{1}{4}$$

□

Exercise 1.2.5. Determine the mean and variance of the sample mean:

$$\overline{X} = 5^{-1}\sum_{i=1}^{5} X_i$$

where, X_i are iid having pdf:

$$f(x) = \begin{cases} 4x^3 & x \in (0,1) \\ 0 & \text{elsewhere} \end{cases}$$

Solution,

Because of an iid sample, we first to figure out the expectation and variance of X.

So,

$$\mathbb{E}X = \int_0^1 \mathrm{d}x\, x(4x^3) = \int_0^1 \mathrm{d}x\, 4x^4 = \frac{4}{5}$$

Similarly,

$$\mathbb{E}X^2 = \int_0^1 \mathrm{d}x\, 4x^5 = \frac{2}{3}$$

Thus,

$$\operatorname{var} X = \mathbb{E}X^2 - (\mathbb{E}X)^2 = \frac{2}{3} - \left(\frac{4}{5}\right)^2 = \frac{2}{75}$$

Thus, by linearity and independence:

$$\mathbb{E}\overline{X} = 5^{-1}\left(5\left(\frac{4}{5}\right)\right) = \frac{4}{5}$$
$$\operatorname{var}\overline{X} = 5^{-1}\left(5\left(\frac{2}{75}\right)\right) = \frac{2}{75}$$

□

Exercise 1.2.6. Let X, Y be random variables with $\mu_1 = 1, \mu_2 = 4, \sigma_1^2 = 4, \sigma_2^2 = 6, \rho = \frac{1}{2}$. Find the mean and variance of the random variable:

$$Z = 3X - 2Y$$

Solution,

By linearity,

$$\mathbb{E}Z = 3\mathbb{E}X - 2\mathbb{E}Y = 3(1) - 2(4) = 3 - 8 = -5$$

Now,

$$\operatorname{var} Z = a^\top \Sigma_{X,Y} a = \begin{pmatrix} 3 \\ -2 \end{pmatrix}^\top \begin{pmatrix} 4 & \frac{1}{2} \\ \frac{1}{2} & 6 \end{pmatrix} \begin{pmatrix} 3 \\ -2 \end{pmatrix} = 54$$

Of course, we could also do this without matrix algebra:

$$\begin{aligned} \operatorname{var} Z &= \operatorname{var}(3X) + \operatorname{var}(2Y) + 2\operatorname{cov}(3X, -2Y) \\ &= 9(4) + 4(6) + 2(3)(-2)\left(\frac{1}{2}\right) \\ &= 9(4) + 4(6) - 2(3) \\ &= 54 \end{aligned}$$

□

Exercise 1.2.7. Let X, Y be independent random variable with means μ_1, μ_2 and variances σ_1^2, σ_2^2. Determine the correlation coefficient of X and $Z = X - Y$ in terms of $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2$

Solution,

Since,

$$\operatorname{cov}(X, X - Y) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}^\top \operatorname{diag}(\sigma_1^2, \sigma_2^2) \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \sigma_1^2$$

Or we could also use:

$$\operatorname{cov} \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \operatorname{diag}(\sigma_1^2, \sigma_2^2) \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}^\top = \begin{pmatrix} \sigma_1^2 & \sigma_1^2 \\ \sigma_1^2 & \sigma_1^2 + \sigma_2^2 \end{pmatrix}$$

Both give,

$$\operatorname{cov}(X, Z) = \sigma_1^2$$

This is because the second one gives the covariance matrix.

Now, correlation coefficient is:

$$\rho = \frac{\operatorname{cov}(X, Z)}{\sigma_1 \sigma_2} = \frac{\sigma_1}{\sigma_2}$$

□

Exercise 1.2.8. Let μ, σ^2 denote the mean and variance of the random variable X. Let $Y = c + bX$, where $b, c \in \mathbb{R}$. Show that the mean and variance of Y are:

$$c + b\mu \quad b^2\sigma^2$$

Solution,

So,

$$\mathbb{E}Y = \mathbb{E}(c + bX) = c + b\mathbb{E}X = c + b\mu$$

Now, variance:

$$\operatorname{var} Y = \operatorname{var}(c + bX) = \operatorname{var}(c) + \operatorname{var}(bX) = b^2 \operatorname{var} X = b^2\sigma^2$$

□

Exercise 1.2.9. Determine the correlation coefficient of the random variables X and Y if $\operatorname{var}(X) = 4$, $\operatorname{var} Y = 2$, and $\operatorname{var}(X + 2Y) = 15$

Solution,

Recall that:

$$\begin{aligned}
\operatorname{var}(X + 2Y) &= \operatorname{var} X + \operatorname{var}(2Y) + 2\operatorname{cov}(X, 2Y) \\
&= \operatorname{var} X + 4\operatorname{var} Y + 4\operatorname{cov}(X, Y) \\
\operatorname{cov}(X, Y) &= \frac{\operatorname{var}(X + 2Y) - \operatorname{var} X - 4\operatorname{var} Y}{4} \\
&= \frac{15 - 4 - 4(2)}{4} \\
&= \frac{3}{4}
\end{aligned}$$

Thus,

$$\rho = \frac{3}{4} \times \frac{1}{\sqrt{4 \times 2}} = \frac{8}{8\sqrt{2}}$$

□

Exercise 1.2.10. Let X and Y be random variables with means μ_1, μ_2; variances σ_1^2, σ_2^2; and correlation coefficient ρ. Show the correlation coefficient of $W = aX + b$, $Z = cY + d$, $a, b \in \mathbb{R}^+$ is ρ.

Solution,

Recall that,

$$\begin{aligned}
\operatorname{cov}(W, Z) &= \mathbb{E}(aX + b - a\mu_1 - b)(cY + d - c\mu_2 - d) \\
&= \mathbb{E}a(X - b)(Y - \mu_2)c \\
&= ac\operatorname{cov}(X, Y)
\end{aligned}$$

Now,

$$\operatorname{var} W = a^2\sigma_1^2 \quad \operatorname{var} Z = c^2\sigma_2^2$$

Thus,

$$\rho_{WZ} = \frac{ac\,\mathrm{cov}(X,y)}{ac\sigma_1\sigma_2} = \frac{\mathrm{cov}(X,Y)}{\sigma_1\sigma_2} = \rho$$

□

Exercise 1.2.11. A person rolls a die, tosses a coin, and draws a card from an ordinary deck. He receives \$3 for each point up on the die, \$10 for a head and \$0 for a tail, and \$1 for each spot on the card (jack = 11, queen = 12, king = 13). If we assume that the three random variables involved are independent and uniformly distributed, compute the mean and variance of the amount to be received.

Solution,

Let (X, Y, Z) give the money received in the experiment.

So, X is a uniform discrete distribution with probability $\frac{1}{6}$ with support $\{3x : x \in \{1, ..., 6\}\}$.

Similarly, Y is a Bernoulli variable with $p = \frac{1}{2}$ with support:

$$\{0, 10\}$$

Finally, Z is a uniform discrete distribution with probability $\frac{4}{52} = \frac{1}{13}$ (this is because we don't care about suits). With support,

$$\{x : x \in \{1, ..., 13\}\}$$

Because we have an independent sample, we first work through thhe mean and variances of all individual variables before computing the mean and variance of the random vector.

Now,

$$\mathbb{E}X = \sum_{x=1}^{6}(3x)\left(\frac{1}{6}\right) = \frac{1}{2}\sum_{x=1}^{6}x = \frac{1}{2}(6)\frac{7}{2} = \frac{21}{2}$$

For variance,

$$\mathbb{E}X^2 = \sum_{x=1}^{6}(3x)^2\left(\frac{1}{6}\right) = \frac{273}{2}$$

Thus,

$$\operatorname{var} X = \frac{273}{2} - \left(\frac{21}{2}\right)^2 = \frac{105}{4}$$

Similarly, we get:

$$\mathbb{E}Y = 5 \quad \operatorname{var} Y = 25$$

and,

$$\mathbb{E}Z = 7 \quad \operatorname{var} Z = 14$$

Here, we're considered with the variable:

$$U = X + Y + Z = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}^\top \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

Now, the mean:

$$\mathbb{E}U = \mathbb{E}X + \mathbb{E}Y + \mathbb{E}Z = \frac{21}{2} + 5 + 7 = \frac{45}{2}$$

Similarly, the variance is also easily computed as:

$$\operatorname{var} U = \operatorname{var} X + \operatorname{var} Y + \operatorname{var} Z = \frac{105}{4} + 25 + 14 = \frac{261}{4}$$

□

Exercise 1.2.12. Let X_1, X_2 be independent random variables with nonzero variances. Find the correlation coefficient of $Y = X_1 X_2$ and X_1 in terms of means and variances of X_1 and X_2.

Solution,

Let $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2$ be the means and variances of X_1, X_2 respectively.

Now,

$$\begin{aligned} \operatorname{cov}(Y, X_1) &= \mathbb{E}(X_1 X_2 - \mu_1 \mu_2)(X_1 - \mu_1) \\ &= \mathbb{E}(X_1^2 X_2 - X_1 X_2 \mu_1 - X_1 \mu_1 \mu_2 + \mu_1^2 \mu_2) \end{aligned}$$

By independence, we can write,

$$\begin{aligned}\operatorname{cov}(Y, X_1) &= \mathbb{E}(X_1^2)\mu_2 - \mu_1^2\mu_2 - \mu_1^2\mu_2 + \mu_1^2\mu_2 \\ &= \mathbb{E}(X_1^2)\mu_2 - \mu_1^2\mu_2 \\ &= \mu_2(\mathbb{E}(X_1^2) - \mu_1^2) \\ &= \sigma_1^2\mu_2\end{aligned}$$

Now,

$$\operatorname{var} Y = \operatorname{var}(X_1X_2) = \mathbb{E}(X_1^2X_2^2) - \mu_1^2\mu_2^2$$

Independence,

$$\operatorname{var} Y = \mathbb{E}(X_1^2)\mathbb{E}(X_2^2) - \mu_1^2\mu_2^2 = \mu_1^2\sigma_2^2 + \mu_2^2\sigma_1^2 + \sigma_1^2\sigma_2^2$$

Thus,

$$\rho_{YX_1} = \frac{\sigma_1^2\mu_2}{\sqrt{(\mu_1^2\sigma_2^2 + \mu_2^2\sigma_1^2 + \sigma_1^2\sigma_2^2)\sigma_1^2}}$$

□

Exercise 1.2.13. Let X_1, X_2, X_3 be random variables with equal variances but with correlation coefficients $\rho_{12} = 0.3, \rho_{13} = 0.5, \rho_{32} = 0.2$. Find the correlation coefficient of the linear function $Y = X_1 + X_2, Z = X_2 + X_3$

Solution,

Here,

$$X = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}$$

Then,

$$Y = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}^\top X \quad Z = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}^\top X$$

Since,

$$\operatorname{cov}(Y, Z) = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}^\top \operatorname{cov}(X) \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

We get:

$$\operatorname{cov}(Y, Z) = \sigma_{12} + \sigma_{13} + \sigma_{22} + \sigma_{23}$$

using the problem statement, and:

$$\sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$$

We get:

$$\operatorname{cov}(Y, Z) = (0.3 + 0.5 + 1 + 0.2)\sigma^2 = 2\sigma^2$$

The correlation coefficient is:

$$\rho = \frac{2\sigma^2}{(2\sigma)(2\sigma)} = \frac{1}{2}$$

□

Exercise 1.2.14. Find the variance of the sum of 10 random variables if each has variance 5 and if each pair has correlation coefficient 0.5

Solution,

Let $\{X_1, ..., X_{10}\}$ be the random variables. We get:

$$\operatorname{var}\left(\sum_{i=1}^{10} X_i\right) = \sum_{i=1}^{10} \operatorname{var}(X_i) + 2\sum_{i<j} \operatorname{cov}(X_i, X_j)$$

By the question,

$$\operatorname{var} X_i = 5 \quad \operatorname{cov}\big(X_i, X_j\big) = (0.5)\sqrt{5 \times 5} = \frac{5}{2}$$

Thus, we get:

$$\operatorname{var}\left(\sum_{i=1}^{10} X_i\right) = 10(5) + 2\left(\frac{5}{2}\right)\frac{(10)(9)}{2} = 275$$

□

Exercise 1.2.15. Let S^2 be the sample variance of the random sample from a distribution with variance $\sigma^2 > 0$. Since $E(S^2) = \sigma^2$ why isn't $E(S) = \sigma$

Solution,

Jensen's inequality immediately gives us:

$$(ES)^2 \leq ES^2$$

This is because x^2 is convex.

However,

$$S = \sqrt{\frac{1}{n-1}\left(\sum_{i=1}^{n} X_i^2 - \frac{\mu^2}{n}\right)}$$

This expression can never vanish for $n \geq 2$:

$$\sum_{i=1}^{n} X_i^2 > \frac{\mu^2}{n}$$

Thus, x^2 here is strictly convex. And so the inequality is strict.

□

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2 Some Special Distribution

2.1 The Binomial and Related Distributions

Definition 2.1.1 (Bernoulli Experiment): A Bernoulli experiment is a random experiment with two mutually exclusive outcomes usually called success or failure.

Remark: The key feature of a Bernoulli is binary outcome that are mutually exclusive.

Definition 2.1.2 (Bernoulli Trials): A sequence of Bernoulli trials occurs when a Bernoulli experiment is performed several independent times so that the probability of success, say p, remains the same from trial to trial.

□

Definition 2.1.3 (Bernoulli Random Variable): X is said to be a Bernoulli random variable with parameter p ($X \sim \text{Bernoulli}(p)$) if:

$$X(\text{success}) = 1 \quad X(\text{failure}) = 0$$

and,

$$\mathbb{P}(X = 1) = p \quad \mathbb{P}(X = 0) = 1 - p$$

The pmf of X is:

$$p(x) = p^x (1 - p)^x$$

□

Theorem 2.1.1 (Mean and Variance of a Bernoulli Variable): Let $X \sim \text{Bernoulli}(p)$, then:

$$\mathbb{E}X = p \quad \text{var}\, X = p(1-p)$$

Proof,

Computing the mgf,

$$M(t) = \sum_{x=0}^{1} e^{tx} p^x (1-p)^{1-x} = q + pe^t$$

where, $q = 1 - p$.

Differentiating, we get:

$$M'(t) = pe^t \quad M''(t) = pe^t$$

Thus,

$$\mathbb{E}X = M'(0) = p \quad \mathbb{E}X^2 = \frac{\mathrm{d}^2 M}{\mathrm{d}t^2}(0) = pe^0 = p$$

Therefore,

$$\text{var}\, X = p - p^2 = p(1-p)$$

And so we're done.

□

Definition 2.1.4 (Binomial Distribution): A discrete random variable X is said to have the Binomial distribution parameterized by n, p if:

$$X = \sum_{i=1}^{n} X_i$$

where, $\{X_i\}$ is an iid sample of Bernoulli variable $X \sim \text{Bernoulli}(p)$. In other words, X is the sum of the successes in a Bernoulli trial of length n.

We notate, $X \sim \text{Binomial}(n, p)$

□

Using this definition, we get the following theorem:

Theorem 2.1.2 (Properties of a Binomial Variable): Let $X \sim \text{Binomial}(n, p)$. Then,

$$p(x) = \begin{cases} \binom{n}{x} p^x q^{n-x} & 0 \leq x \leq n \\ 0 & \text{otherwse} \end{cases}$$

where, $q = 1 - p$. Then, p is the pmf of X.

Moreover,

$$M(t) = \left(q + pe^t\right)^n \quad \mathbb{E}X = np \quad \text{var}\, X = npq$$

Proof,

Since, X is the sum of idd random variables X_i, its mgf is given by:

$$M(t) = \left[M_{X_i}(t)\right]^n$$

where, $M_{X_i}(t)$ is the mgf of X_i.

Recall that,

$$M_{X_i}(t) = q + pe^t$$

Thus,

$$M(t) = \left(q + pe^t\right)^n$$

By binomial theorem, we get:

$$M(t) = \sum_{x=0}^{n} \binom{n}{x} q^{n-x} p^x e^{tx}$$

Using the fact that:

$$M(t) = \sum_{x=-\infty}^{\infty} p(x) e^{tx}$$

By uniqueness of mgf,

$$p(x) = \begin{cases} \binom{n}{x} p^x q^{n-x} & 0 \leq x \leq n \\ 0 & \text{otherwse} \end{cases}$$

Finally, to get moments, we use our mgf:

$$\frac{\mathrm{d}M}{\mathrm{d}t} = npe^t\left(q + pe^t\right)^{n-1}$$

$$\frac{\mathrm{d}^2M}{\mathrm{d}t^2} = npe^t(q + npe^t)\left(q + pe^t\right)^{n-2}$$

Substituting $t = 0$ and using $p + q = 1$, we get:

$$\frac{\mathrm{d}M}{\mathrm{d}t} = np \quad \frac{\mathrm{d}^2M}{\mathrm{d}t^2} = np(q + np)$$

So,

$$\mathbb{E}X = np \quad \mathbb{E}X^2 = np(q + np)$$

Finally,

$$\operatorname{var} X = np(q + np) - n^2p^2 = npq$$

□

Theorem 2.1.3 (Additive Property Binomial Variables): Let $X_1, ..., X_m$ be independent random variable such that $X_i \sim \text{Binomial}(n_i, p)$. If,

$$Y = \sum_{i=1}^{m} X_i$$

. Then,

$$Y \sim \text{Binomial}\left(\sum_{i=1}^{m} n_i, p\right)$$

Proof,

Let,

$$M_i(t) = \left(q + pe^t\right)^{n_i}$$

be the mgf of X_i.

By independence, it follows that the mgf of Y satisfies:

$$M(t) = \prod_{i=1}^{m} M_i(t) = \left(q + pe^t\right)^n$$

where,

$$n = \sum_{i=1}^{m} n_i$$

By uniqueness of mgf, we recognize that:

$$Y \sim \text{Binomial}(n, p)$$

□

Now, we move to distributions that are related to the binomial distribution.

□

Definition 2.1.5 (Negative Binomial Distribution): $X \sim \text{NegBin}(r, p)$ if:

$$X(r \text{ successes at the } x\text{th trial}) = x$$

Remark: There is an equivalent formulation of the negative binomial distribution. Here, instead of counting successes, we count failures.

□

Theorem 2.1.4 (PMF of $\text{NegBin}(r, n)$): Let $X \sim \text{NegBin}(x, r)$, then:

$$p(x) = \begin{cases} \binom{x-1}{r-1} p^r q^{x-r} & x \geq r \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

Proof,

Let,

$$A = r - 1 \text{ successes before } x - 1 \text{ trials}$$
$$B = \text{success on the trial } x$$

Then,

$$A \cap B = r\text{th success on } x\text{th trial}$$

By multiplication formula for probabilities,

$$p(x) = \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A)$$

By independence,

$$\mathbb{P}(B \mid A) = \mathbb{P}(B)$$

Thus,

$$p(x) = \mathbb{P}(A)\mathbb{P}(B)$$

The binomial connection happens for computing $\mathbb{P}(A)$. Since if we're given $Y \sim \text{Bin}(x-1, p)$, then,

$$\mathbb{P}(Y = r-1) = \mathbb{P}(A) = \binom{x-1}{r-1} p^{r-1} q^{x-r}$$

$\mathbb{P}(B) = p$ by definition, so:

$$p(x) = \binom{x-1}{r-1} p^{r} q^{x-r}$$

Of course, here we have supposed that $x \in \mathcal{S}_X$.

Computing $\mathcal{S}_X$ is also easy $p(x)$ only vanishes if and only if $\mathbb{P}(A)$ vanishes. Recall the suppport of the binomial distribution and we're done.

□
